ATTENDANCE RECOVERY COMPUTATION

Python Project Report

1.Overview

Attendance Recovery Computation is a Python-based project designed to calculate student attendance percentages and determine the number of additional classes required to reach the minimum attendance requirement. It helps students monitor their attendance and plan their attendance recovery efficiently.

2.Example Record Format

Student ID	Student Name	Total Classes	Attended Classes	Attendance Percentage
S101	Ravi	100	65	65%
S102	Priya	100	80	80%
S103	Kumar	100	90	90%

3.Problem Statement

Students often find it difficult to calculate their attendance percentage and determine how many additional classes they need to attend to meet the minimum attendance requirement. This project automates attendance percentage calculation and estimates the number of consecutive classes required to reach the target attendance percentage.

4.Objectives

1. To calculate student attendance percentages.
2. To identify students who have low attendance.
3. To determine the additional classes required to reach the target attendance percentage.
4. To provide accurate attendance recovery calculations.
5. To help students monitor and improve their attendance.

5.Assumptions

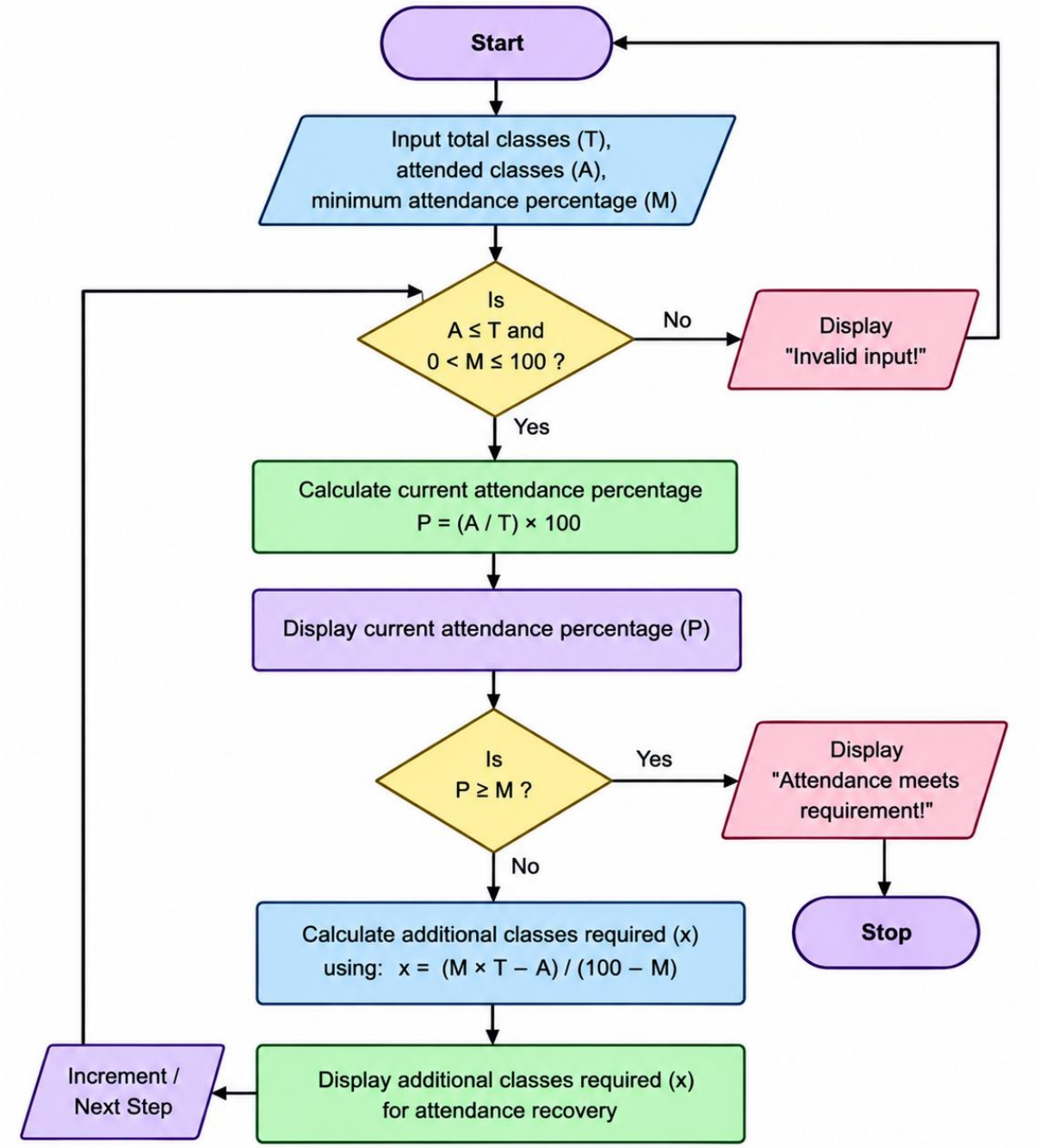
1. The total number of classes and attended classes are provided by the user.
2. The number of attended classes cannot exceed the total classes.
3. The minimum required attendance percentage is predefined.

4. Future classes are assumed to be attended without any absences.
5. Attendance calculations are based on the total number of classes conducted.

6.Algorithm

1. Start the program.
2. Input the total number of classes conducted.
3. Input the number of classes attended.
4. Input the minimum required attendance percentage.
5. Validate the input values.
6. Calculate the current attendance percentage.
7. Check whether the attendance meets the required percentage.
8. If attendance is below the target, calculate the additional consecutive classes required.
9. Display the current attendance percentage and recovery details.
10. Stop the program.

7.Flowchart



8.Pseudocode

BEGIN

INPUT total_classes (T)

INPUT attended_classes (A)

INPUT minimum_attendance_percentage (M)

```
  IF T <= 0 OR A < 0 OR A > T OR M <= 0 OR M > 100 THEN
    PRINT "Invalid input"

  ELSE
    attendance_percentage = (A / T) * 100

    PRINT "Current Attendance Percentage = ", attendance_percentage

    IF attendance_percentage >= M THEN
      PRINT "Attendance meets the requirement"
      PRINT "No additional classes required"

    ELSE
      additional_classes = CEILING((M * T - 100 * A) / (100 - M))

      PRINT "Additional Classes Required = ", additional_classes
    END IF

  END IF

END
```

9.Source Code

```
import math

print("====================================")
print("   ATTENDANCE RECOVERY COMPUTATION")
print("====================================")
```

```
try:
    total_classes = int(input("Enter total number of classes: "))
    attended_classes = int(input("Enter number of classes attended: "))
    minimum_percentage = float(input("Enter minimum attendance percentage: "))

    if total_classes <= 0:
        print("Error: Total classes must be greater than zero.")

    elif attended_classes < 0 or attended_classes > total_classes:
        print("Error: Attended classes must be between 0 and total classes.")

    elif minimum_percentage <= 0 or minimum_percentage > 100:
        print("Error: Minimum percentage must be between 0 and 100.")

    else:
        attendance_percentage = (
            attended_classes / total_classes
        ) * 100

        print("\n---------- ATTENDANCE REPORT ----------")
        print("Total Classes      :", total_classes)
        print("Classes Attended   :", attended_classes)
        print("Current Attendance  : {:.2f}%".format(
            attendance_percentage
        ))
        print("Minimum Required    : {:.2f}%".format(
            minimum_percentage
```

```
))

if attendance_percentage >= minimum_percentage:
   print("\nStatus: Attendance requirement is satisfied.")
   print("Additional Classes Required: 0")

elif minimum_percentage == 100:
   print("\nStatus: Attendance requirement is not satisfied.")
   print("100% attendance cannot be reached by attending")
   print("future classes alone if any classes were missed.")

else:
   numerator = (
      minimum_percentage * total_classes
      - 100 * attended_classes
   )
   denominator = 100 - minimum_percentage

   additional_classes = math.ceil(
      numerator / denominator
   )

   print("\nStatus: Attendance is below the requirement.")
   print("Additional Classes Required:",
       additional_classes)

print("----------------------------------------")
```

```
except ValueError:
    print("Error: Please enter valid numeric values.")
```

10.Test Cases

Test Case	Total Classes	Attended Classes	Minimum Attendance (%)	Expected Output
1	100	60	75	Attendance = 60%. Additional classes required = 60.
2	100	80	75	Attendance = 80%. Requirement satisfied. Additional classes = 0.
3	50	20	75	Attendance = 40%. Additional classes required = 70.
4	100	90	75	Attendance = 90%. Requirement satisfied. Additional classes = 0.
5	0	0	75	Error: Total classes must be greater than zero.
6	100	110	75	Error: Attended classes must be between 0 and total classes.
7	100	60	0	Error: Minimum percentage must be between 0 and 100.
8	100	60	100	Attendance requirement cannot be reached by attending future classes alone because some classes were missed.

Result

The test cases verify that the program correctly calculates the current attendance percentage, checks the minimum attendance requirement, calculates the additional classes required, and handles invalid inputs.

11.Output Screenshots

```
====================================
   ATTENDANCE RECOVERY COMPUTATION
====================================
Enter total number of classes: 100
Enter number of classes attended: 60
Enter minimum attendance percentage: 75

---------- ATTENDANCE REPORT ----------
Total Classes       : 100
Classes Attended    : 60
Current Attendance  : 60.00%
Minimum Required    : 75.00%

Status: Attendance is below the requirement.
Additional Classes Required: 60
----------------------------------------
```

12.Conclusion

The Attendance Recovery Computation project was developed using Python to calculate students' attendance percentages and determine the additional classes required to meet the minimum attendance requirement. The program provides accurate results, identifies students with low attendance, and handles invalid inputs. This project helps students monitor their attendance and plan their future classes effectively. Thus, the project demonstrates the practical application of Python programming in solving real-world problems.